CSE220 Digital Audio Equalizer -- DSP Analysis Report

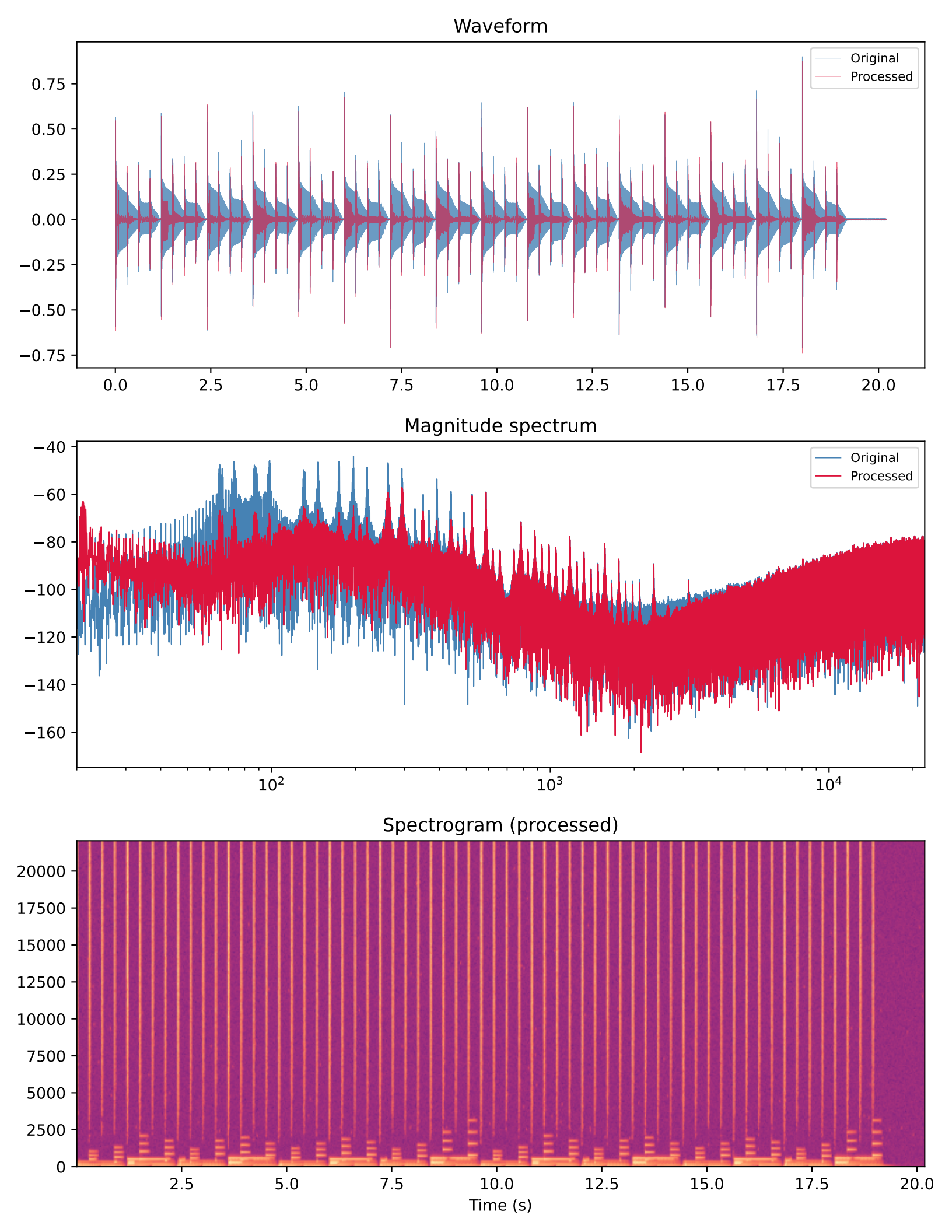
```
Mode: graphic
EQ settings: smoke test

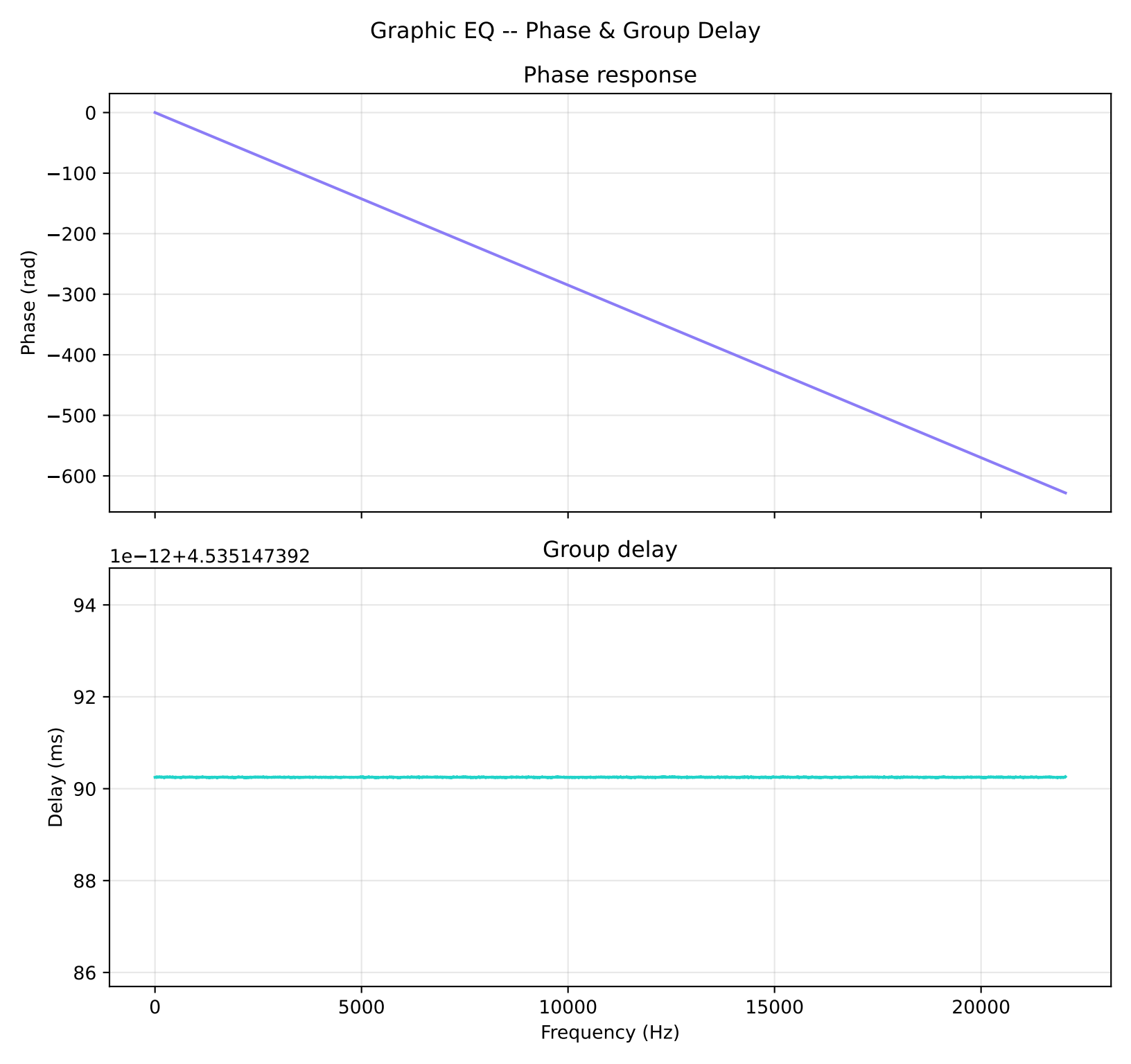
Sample rate: 44100 Hz
Duration: 20.20 s

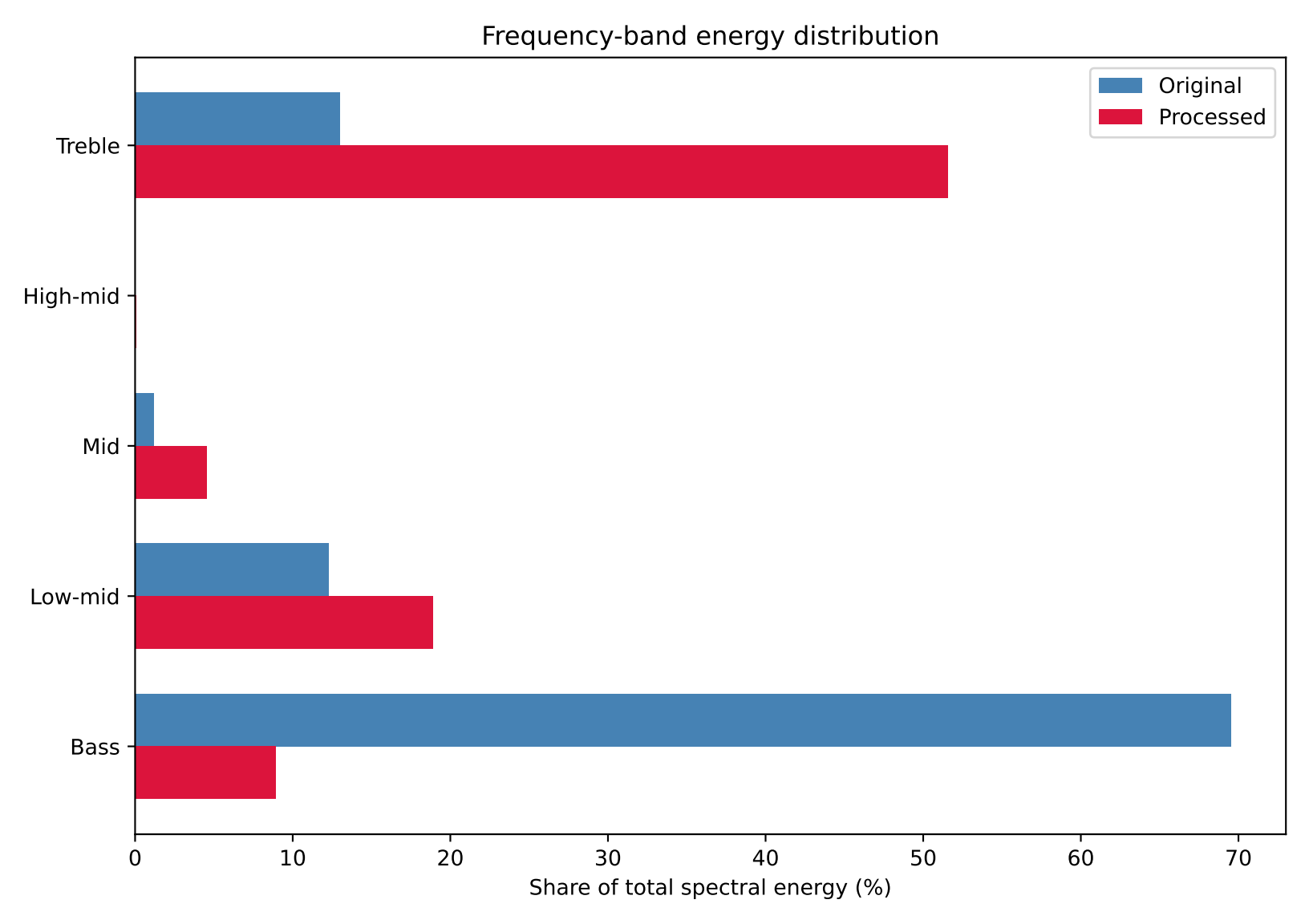
Clipping protection:
  Peak before normalization : 0.8731
  Peak after normalization  : 0.8731
  Clipping detected         : False

Quantitative metrics (Original vs Processed):

Metric                         Original     Processed
-----------------------------------------------------
Duration (s)                      20.20         20.20
RMS                               0.0600        0.0301
Peak                              0.9000        0.8731
Energy                         3210.88        807.98
Spectral Centroid (Hz)         13042.3       14022.2
Dominant Freq (Hz)                196.7         293.3
Zero-Crossing Rate                0.1039        0.1212
```


Waveform
Original
Processed
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Magnitude spectrum
Original
Processed
−40
−60
−80
−100
−120
−140
−160
10^2
10^3
10^4
Spectrogram (processed)
20000
17500
15000
12500
10000
7500
5000
2500
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Graphic EQ -- Phase & Group Delay
Phase response
Phase (rad)
0
−100
−200
−300
−400
−500
−600
Group delay
1e−12+4.535147392
Delay (ms)
94
92
90
88
86
0
5000
10000
15000
20000
Frequency (Hz)


Frequency-band energy distribution
Original
Processed
Treble
High-mid
Mid
Low-mid
Bass
0
10
20
30
40
50
60
70
Share of total spectral energy (%)

Filter Performance & FIR vs IIR

FIR vs IIR comparison:

Property	FIR (Graphic EQ)	IIR (Parametric EQ)
Structure	Parallel bank of 10 band-pass FIR filters	Cascade of 2 biquad section(s)
Stability	Inherently stable (no poles)	Stable (max |pole|=0.9791)
Phase	Linear (symmetric kernel)	Nonlinear (typical for IIR)
Coefficients	401	10
Processing time	469.93 ms	8.71 ms